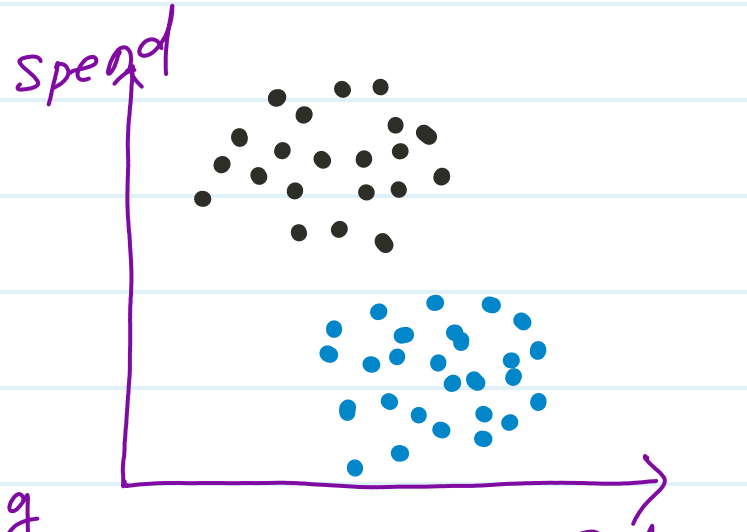
Unsupervised Learning

It doesnot required labe data.

It can be work on structure and unstructure data.

1. K-mean clustering
2. Hierarchical clustering
3. DBSCAN clustering
4. PCA
5. Apriori



Income	Spendin
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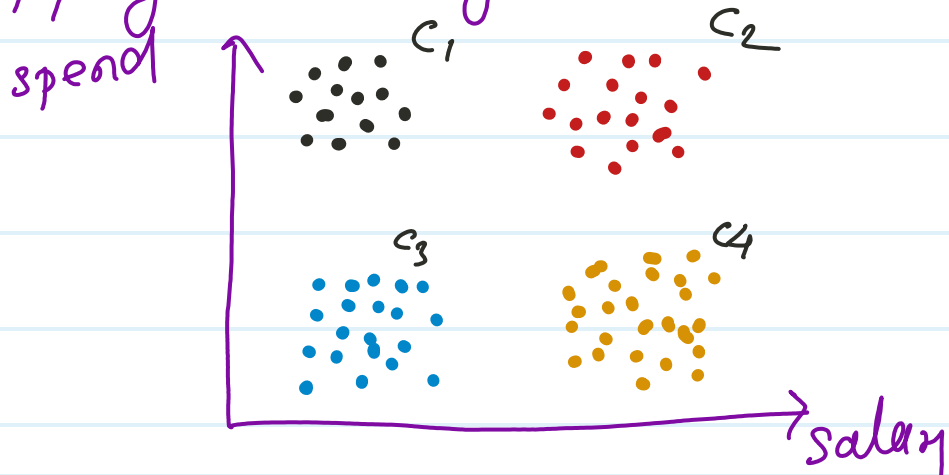
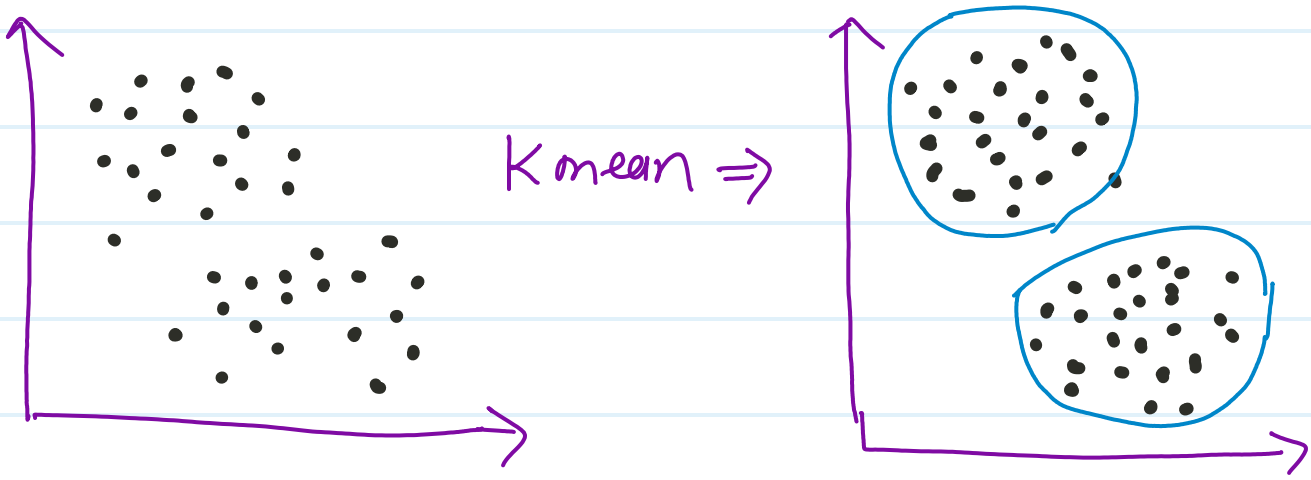
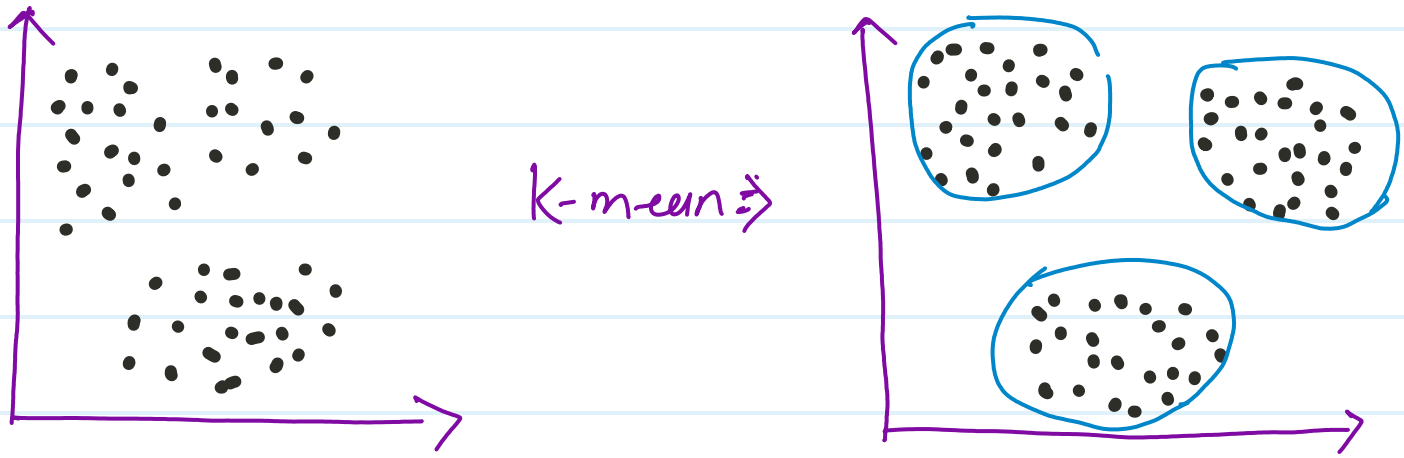
After Applying clustering



K-mean clustering

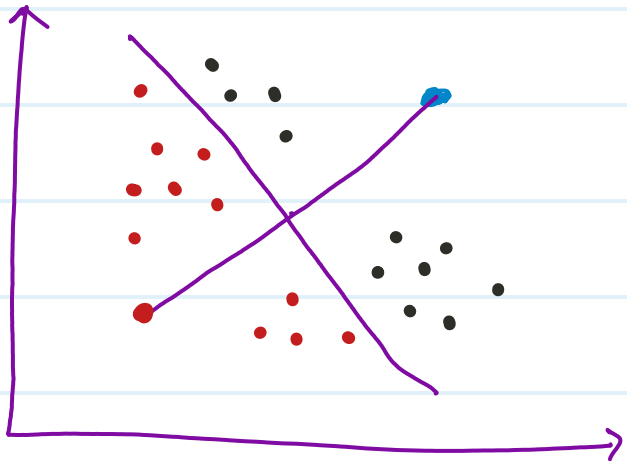
Geometric Intution :-





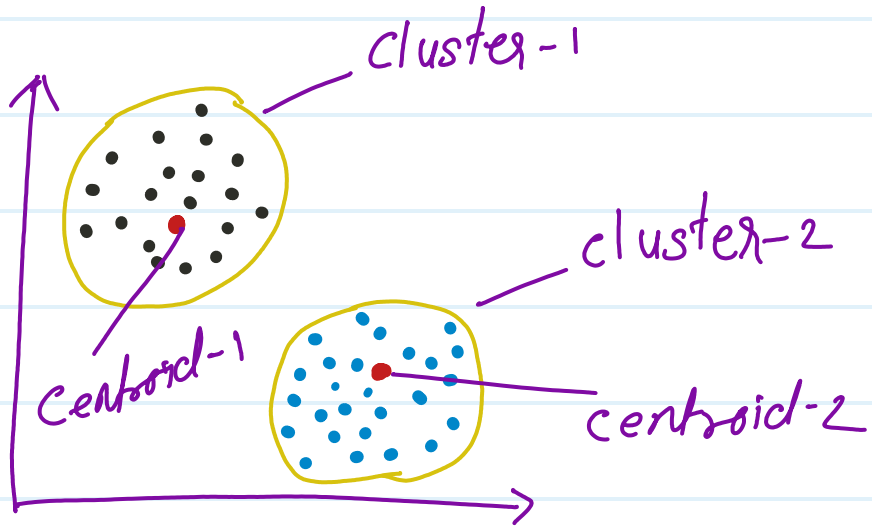
steps -

1. Initialize some new (K) centroid
2. point that are nearest to the centroid group them.
3. move centroid by calculating the mean of points



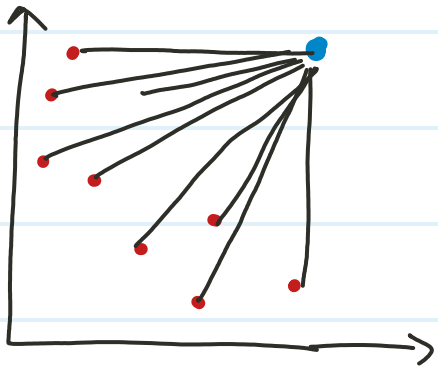
centroid point will sift toward.....

Final output



How to select k-value?

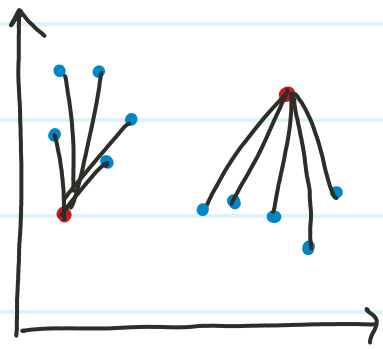
WCSS – within cluster sum of square

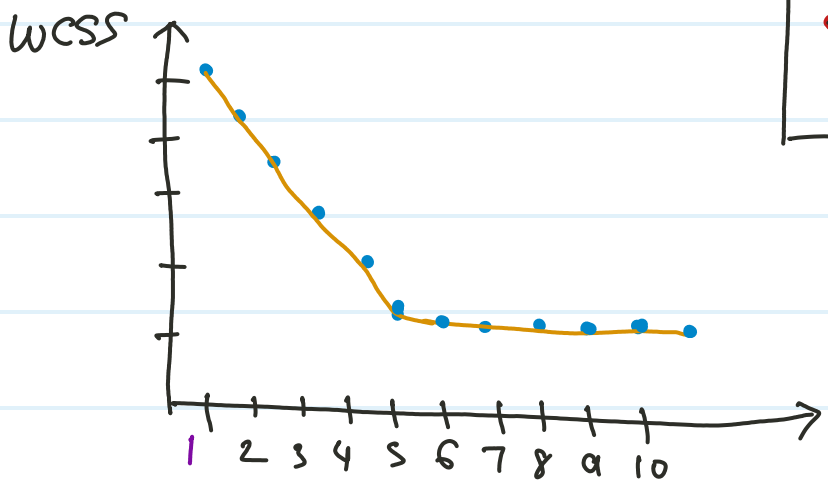


$$WCSS = \sum_{i=0}^{k} \left(\text{Distance b/w point to nearest Centroid} \right)^2$$

suppose $k = 1 - 20$

* Elbo method –

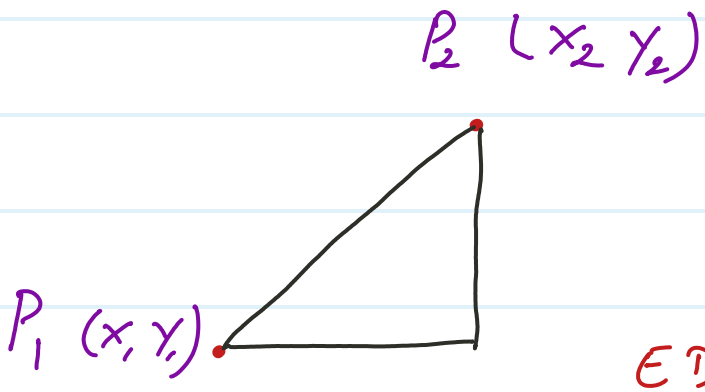




when wcss calculate with $k=1$, wcss value is high, when we increase $k=2$, it will reduce, when $k=3$ it will reduce. when $k=3$ it will reduce and after

Some point it will be straight in graph it is called elbo method.
When curv become straight at that point we got k value. like above graph, 5 will be our k value.

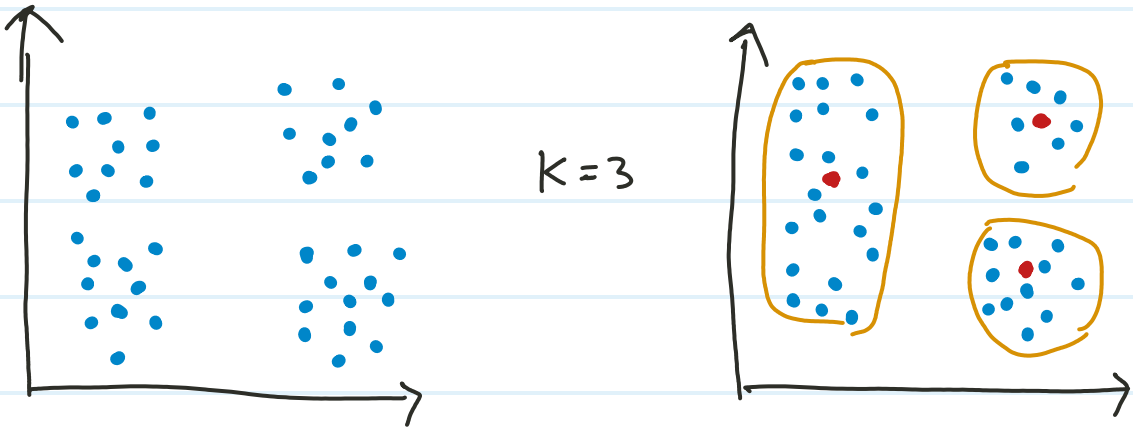
To calculate distance we need.
euclidean distance or Manhattan distanc.

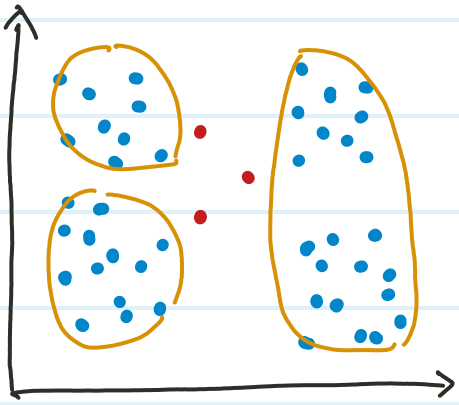


$$ED = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

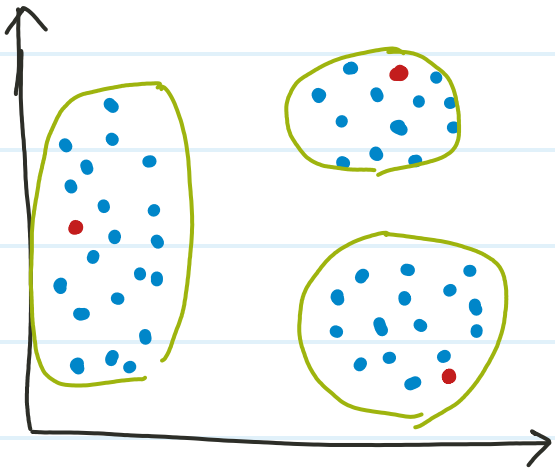
$$MD = |x_2 - x_1| + |y_2 - y_1|$$

Random Initialization trap (K-mean ++)





When centroin form very nearest to each other it stuck in Random trap.

To handle this kind of trap we use K-mean ++ method.

With help of it centroid form far from each other. and prevent from random initialization trap.